

IP address query

- *ifconfig eth0*
 -a
- *ip addr show dev eth0*

IP address setup/modification

- *ifconfig eth0 1.2.3.4 netmask 255.255.0.0 broadcast 1.2.255.255 up*

Compact form:

ifconfig eth0 1.2.3.4/16 broadcast 1.2.255.255 up

ifconfig eth0 1.2.3.4/16

- *ifconfig eth0 down*
up
- *ip addr add 1.2.3.4/16 dev eth0*
del

Dynamic IP address configuration

- *dhclient eth0*

Requests valid IP parameters from the DHCP server available on the network.

Switching the interface on/off

- *ifconfig eth0 **down***
up
- *ip link set dev eth0 **down***
up

Subnetting

Splitting an IP network into subnets:

- We take *m* of the node identification bits and use them for subnet identification. At least *1 bit* must be taken to form subnets, and a maximum number of *node bits - 2 bits* can be used.

Subnetting (example)

Address of the initial network:

192.168.1.0/24	11000000.10101000.00000001.00000000
255.255.255.0	11111111.11111111.11111111.00000000

Subnets

192.168.1.0/25	11000000.10101000.00000001.00000000
255.255.255.128	11111111.11111111.11111111.10000000

192.168.1.128/25	11000000.10101000.00000001.10000000
255.255.255.128	11111111.11111111.11111111.10000000

Subnet domains

192.168.1.0/25 – 192.168.1.127/25

192.168.1.128/25 – 192.168.1.255/25

Subnetting (example)

Address of the initial network:

192.168.1.0/24	11000000.10101000.00000001.00000000
255.255.255.0	11111111.11111111.11111111.00000000

Subnets

192.168.1.0/26	11000000.10101000.00000001.00000000
255.255.255.192	11111111.11111111.11111111.11000000

192.168.1.64/26	11000000.10101000.00000001.01000000
255.255.255.192	11111111.11111111.11111111.11000000

192.168.1.128/26	11000000.10101000.00000001.10000000
255.255.255.192	11111111.11111111.11111111.11000000

192.168.1.192/26	11000000.10101000.00000001.11000000
255.255.255.192	11111111.11111111.11111111.11000000

Subnet domains

192.168.1.0/26 – 192.168.1.63/26
192.168.1.64/26 – 192.168.1.127/26
192.168.1.128/26 – 192.168.1.191/26
192.168.1.192/26 – 192.168.1.255/26

Subnetting (tasks)

Split the 79.0.0.0/8 network into 4 equal subnets! Split the first subnet into 3 additional subnets according to the following needs:

- A./ 90 node
- B./ 200 node
- C./ 500 node

Record the allocated address ranges in a CIDR-compatible format.

Checking IP communication (ping)

The ping command is used to check the IP connection. ICMP (Internet Control Message Protocol) sends out ECHO REQUEST packets, to which the called node responds with an ICMP ECHO REPLY message.

```
ping [options] {ip_address/domain_name}
```

```
ping 172.22.206.1
```

```
ping -c 4 www.c3.hu
```

```
ping -w 5 172.22.206.1
```

Options:

-c {number}: the program stops after sending a given number of ECHO REQUEST requests

-w {time duration}: specify timeout in seconds

-f : flooding (continuous) ping

ping (exercise)

- Check if the neighbouring host is available.
- Ping the **www.c3.hu** server four times, then interpret the output of the command.

```
[root@erlang ~]# ping -c4 www.c3.hu
PING memex.c3.hu (194.38.96.78) 56(84) bytes of data.
64 bytes from memex.c3.hu (194.38.96.78): icmp_seq=1 ttl=54 time=5.02 ms
64 bytes from memex.c3.hu (194.38.96.78): icmp_seq=2 ttl=54 time=5.02 ms
64 bytes from memex.c3.hu (194.38.96.78): icmp_seq=3 ttl=54 time=5.07 ms
64 bytes from memex.c3.hu (194.38.96.78): icmp_seq=4 ttl=54 time=5.05 ms

--- memex.c3.hu ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 5.028/5.047/5.072/0.019 ms
```

56(84): ICMP message size (IP packet size) in bytes

icmp_seq: echo reply response to an echo request with the specified number of lines

ttl: time-to-live (determine how many routers the response of the called node came back to us through.)

time: response time or round trip time

Analysis of a **statistical block**.